

Key features

Feature	Ceph	MinIO
Storage interfaces	Object (S3/Swift via RGW), Block (RBD), File (CephFS)	Object (S3 API) only
Data protection	Replication or erasure coding, configurable per pool	Erasure coding (Reed-Solomon), per-bucket
Scale	Tens of PB, thousands of nodes	Scales via distributed mode across nodes/drives
Self-healing	Automatic rebalancing and recovery	Automatic bitrot detection and healing (mc admin heal)
Multi-tenancy	CephX authentication, pools, namespaces	IAM-style policies, multi-tenant buckets
Encryption	At-rest (dm-crypt), in-transit (msgr2)	SSE-S3, SSE-C, SSE-KMS support
Tiering	CRUSH-based placement rules across device classes	Integrated tiering to S3-compatible cold storage
Monitoring	Ceph Dashboard, Prometheus exporter	Built-in Prometheus metrics, MinIO Console

- Raw block devices for OSDs (Object Storage Daemons) — do not use pre-partitioned/formatted disks.
- Container runtime (Podman or Docker) — modern Ceph deploys via cephadm, which runs services in containers.
- Time sync across nodes (chrony/NTP) — clock drift breaks cluster quorum.
A dedicated storage/cluster network is strongly recommended in addition to the public network.
To install MinIO, you require:
 - Linux, Windows, or macOS host(s); production deployments should use Linux.
 - Raw or XFS-formatted drives (XFS recommended for MinIO's erasure-coded backend).
 - At least four drives per node for distributed erasure coding (minimum recommended for testing).
 - mc (MinIO Client) for administration and bucket management.
 - Open ports: 9000 (API) and 9001 (Console) by default.

Installation

You can install Ceph (via cephadm, single-node lab cluster) as follows:

```
# Install prerequisites
sudo apt update && sudo apt install -y podman chrony curl
```

```
# Download and install cephadm
curl --silent --remote-name --location \
  https://download.ceph.com/rpm-18.2.0/el9/noarch/cephadm
chmod +x cephadm
sudo ./cephadm add-repo --release reef
sudo ./cephadm install
```

```
# Bootstrap the cluster (run on the first/admin node)
sudo cephadm bootstrap --mon-ip <MON_IP_ADDRESS>
```

```
# Add OSDs from all available raw devices
sudo ceph orch apply osd --all-available-devices
```

```
# Enable the RGW (S3-compatible) service
sudo ceph orch apply rgw myrealm myzone --placement="1
<HOSTNAME>"
```

The bootstrap step outputs a dashboard URL, admin credentials, and a *ceph.conf* plus keyring you'll use for subsequent ceph CLI commands.

To install MinIO (single-node, distributed-ready), use the following code:

```
# Download the MinIO server binary
wget https://dl.min.io/server/minio/release/linux-amd64/minio
chmod +x minio
sudo mv minio /usr/local/bin/
```

```
# Create data directories (example: 4 drives)
sudo mkdir -p /mnt/disk{1,2,3,4}/minio
```

```
# Install the MinIO Client (mc)
wget https://dl.min.io/client/mc/release/linux-amd64/mc
chmod +x mc
sudo mv mc /usr/local/bin/
```

For a multi-node distributed deployment, MinIO is started identically on each node, pointing at the full set of endpoints:

```
minio server http://node{1...4}.example.com:9000/mnt/
disk{1...4}/minio \
  --console-address ":9001"
```

Configuration of environment variables

MinIO is configured primarily through environment variables, which is convenient for containerized and Kubernetes deployments.

```
# Root credentials (required)
```